

Optimization and design to catalyze sustainable energy in Morocco's Eastern Sahara: A hybrid energy system of PV/Wind/PHS for rural electrification

Norddine Oubouch^{*}, Abdelbari Redouane, Anouar Makhoukh, Abdennebi El Hasnaoui

Laboratoire Commande, Pilotage et Surveillance en Energie Electriques (CPS2E), Ecole nationale supérieure des mines de Rabat, Rabat, 10090, Morocco

ARTICLE INFO

Keywords:

HOMER
Hybrid energy system
Morocco
Optimization
Pumped hydro storage
Techno-economic

ABSTRACT

This paper conducts a comprehensive assessment of the potential of water, solar, and wind resources for sustainable energy generation. The study is situated in a Moroccan region within eastern Saharan Africa. It presents a detailed comparative analysis between a photovoltaic system (PV) integrated with a pumped hydro storage (PHS), a wind turbine, and a conventional grid, considering both energy production and economic analysis using HOMER software. Moreover, the paper provides an initial social impact assessment of hybrid energy systems integrating locally available water resources, especially during the winter season, alongside photovoltaic and wind technologies. This evaluation delves into aspects of rural electrification and community development. The findings underscore the potential of sustainable energy solutions to drive economic and social progress in the studied area by harnessing the region's water resources. We proposed this technology because the owners of the area do not greatly benefit from the seasonal groundwater that passes through the valley, despite the presence of a dam. Accordingly, we will exploit this water to generate energy and achieve energy self-sufficiency. By harnessing this underutilized resource, we aim to provide sustainable energy solutions and drive economic and social progress in the region. The results given by HOMER identify the most cost-effective system capable of serving the load at the lowest cost of energy (COE) of about \$0.03831 and net present cost (NPC) of about \$262,596 under the modeled conditions, and the most satisfactory system chosen by the HOMER optimizer is a PV/Wind/PHS-based hybrid energy system.

1. Introduction

In recent years, there has been a notable global uptake of renewable energy sources, particularly wind, photovoltaic (PV) systems, and pumped hydro energy storage. These sources are valued for their abundance and eco-friendliness but face challenges due to climate dependency. In the present era, marked by acute environmental consciousness and a commitment to sustainable energy practices, it becomes increasingly clear that harnessing both solar and wind energy simultaneously holds significant promise. Integrating pumped hydro-power can significantly boost system performance and efficiency. This synergy ensures reliable power and flexibility to meet increasing energy demands. Previous studies have explored hybrid renewable energy systems to electrify rural areas. Hou et al. and Wimalaratna et al. collectively studied advanced renewable energy solutions, optimizing wind-photovoltaic-storage systems, assessing wind power integration, and introducing an innovative off-grid system for sustainable energy generation. They utilized the Cat Swarm Optimization approach, which yielded considerable optimal results. (Hou et al., 2020; Wimalaratna et al., 2022). A. Allouhi harnessed a combination of solar photovoltaic

panels, wind turbines, and micro-hydropower systems, which was the most effective configuration to engineer an optimal energy solution for rural Morocco (Allouhi, 2024). Ennemiri et al. evaluate a hybrid solar PV-biogas-battery system, showcasing its cost-effectiveness, resilience, and a significant reduction in CO₂ emissions compared to single-source systems (Ennemiri et al., 2024). Guezgouz et al. used Matlab to evaluate the economic feasibility of a hybrid energy storage system. They found that combining pumped water storage with batteries, solar, and wind energy is optimal for long-term provision compared to using batteries alone (Guezgouz et al., 2019). Al-Masri et al. proposed optimal cost-effective scenarios for a hybrid energy system, minimizing grid dependence through algorithmic improvements. The identified optimal system integrates solar energy, wind, and pumped hydro storage (Al-Masri et al., 2023). Javed et al. investigated the viability of combining pumped hydro and battery storage for renewable energy-powered systems. Their research highlights hybrid storage systems' enhanced reliability and cost-effectiveness, contributing to more efficient renewable energy utilization (Javed et al., 2020). Kamar et al. proposed a system exceeding hybrid energy system standards, optimizing through

^{*} Corresponding author.

E-mail address: nourddin.oubouch@gmail.com (N. Oubouch).

<https://doi.org/10.1016/j.cles.2024.100141>

Received 6 February 2024; Received in revised form 4 August 2024; Accepted 20 August 2024

Available online 28 August 2024

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algorithms for economic and social feasibility with cost-effective and premium solutions based on seasonal forecasting (Kumar et al., 2023). Bhandari et al. examine innovative hybridization methods for off-grid systems, integrating solar PV, wind, and micro-hydro, fostering environmental sustainability and global replication (Bhandari et al., 2014). Roman et al. examined a hybrid system's economic and energy aspects to electrify a rural area, testing three battery types. They concluded that integrating solar energy, diesel generators, and absorbent glass mat batteries is the most optimal and economically feasible (Román et al., 2022). Sharma et al. modeled and analyzed an economically favorable grid-connected hybrid green microgrid system with promising implementation opportunities (Sharma et al., 2022). Ma et al.'s study demonstrates the technical feasibility of combining pumped hydro storage with a standalone solar-wind system to ensure a reliable and eco-friendly energy supply for remote communities (Ma et al., 2014). Nyeche et al. propose a hybrid energy conversion technology integrating solar PV arrays, wind turbines, and a pumped hydro energy storage (PHES) system, employing HOMER Pro and MATLAB for genetic algorithms and economic models (Nyeche and Diemuodeke, 2020). Ramesh et al. utilized HOMER program strategies (Cycle Charging and Load Following) and compared two batteries to identify the optimal system's compatibility with a hybrid energy system incorporating solar, wind, hydro pumps, and diesel generators (Ramesh and Saini, 2020).utilized HOMER software to design an optimized hybrid renewable energy system that combines solar PV, wind, and storage. Their study showcases significant reductions in pollution, gas emissions, and operational costs (Khalil et al., 2021). Manas et al. in this study present an isolated microgrid integrating wind, solar, biogas, and battery, managed by a discreet algorithm for load-supply balance, enhancing energy sustainability in diverse conditions (Manas et al., 2015). Rad et al. propose an economic hybrid system of solar, wind, and biogas for cost-effective electricity supply to a remote village. Integrating fuel enhances flexibility while also increasing environmental pollution (Rad et al., 2020). Kousksou et al.'s paper underscores Morocco's considerable renewable energy potential. It outlines the national strategy for a sustained low-carbon economy, addressing challenges posed by population and economic growth (Kousksou et al., 2015). Ashraf et al. proposed an optimized hybrid solar/diesel/battery system e-constraint in desert areas using advanced optimization techniques, demonstrating superior cost-effectiveness and reduced environmental impact compared to traditional methods (Ashraf et al., 2020). Oubouch et al. and Shahzad et al. delved into the realm of hybrid energy systems tailored for remote village electrification. Their investigation focused on pinpointing the optimal synergy between solar energy and biomass resources. Through meticulous sensitivity analysis, they discerned the most financially prudent and economically viable strategies (Oubouch et al., 2023; Shahzad et al., 2017). Redouane et al. and Benchraa et al. investigate hybrid energy systems combining biomass, PV, and batteries with different grid connection scenarios. They focus on combined heat and power (CHP) options, such as gas-fired, diesel-powered, and wood-burning HP, aiming to fulfill residents' energy needs and promote economic benefits (Redouane et al., 2015; Benchraa et al., 2018). Esen et al. collectively conducted studies evaluating the economic feasibility of a ground source heat pump (GSHP) for heating and performing a techno-economic comparison of ground-coupled and air-coupled heat pump systems for space cooling (Esen et al., 2006, 2007). Esen and Yuksel explored greenhouse heating with renewables, highlighting biogas and slinky-type heat exchangers' effectiveness, and proposing integrated agricultural sustainability systems (Esen and Yuksel, 2013). Krishnamoorthy et al. assess a solar PV array-powered water purification plant's cost-effectiveness and advocate efficient energy storage for rural electrification, favoring the PV/BG-based hybrid system with lithium-ion batteries for its economic and environmental benefits, demonstrating a commitment to sustainability (Krishnamoorthy et al., 2023, 2022).

Based on the state-of-the-art review, a hybrid system combining photovoltaic (PV) panels, wind turbines, and pumped hydro storage (PHS) can ensure energy access in stand-alone configurations and improve the reliability of weak grids. In the context of fragile rural communities in Saharan Africa, it is crucial to analyze economic feasibility and environmental benefits. Our study introduces an innovative use of harmful seasonal floodwaters for hermetic pumped hydro storage, leveraging this resource to store energy and ensure a steady power supply. This approach is preferred over battery storage or systems relying solely on wind or PV. Utilizing available water enhances the cost-effectiveness of PHS in the region. Focusing on Morocco's eastern Sahara, this study aims to achieve energy self-sufficiency, promote economic and social development, and provide new practical solutions for sustainable rural electrification, potentially benefiting the broader region.

This paper follows a structured outline, as summarized below: Section 1 elaborates on the study's context, providing an introduction. Section 2 delves into the methodology, offering insights into data collection and baseline evaluation, a comprehensive viability analysis, hybrid system configuration, and the geographical area. Moving to Section 3 provides detailed explanations of the mathematical and economic aspects of each component. Section 4 delivers an assessment of village resources and energy demand. The outcomes of simulations, optimization, and the study's techno-economic aspects are highlighted in Section 5. Lastly, Section 6 supplies a concise summary of the main conclusions derived from the study.

2. Methodology

This comprehensive project encompasses a multifaceted approach, integrating rigorous assessments, strategic directives, and thorough economic analyses that delve into the details of Net Present Cost (NPC) and Cost of Energy (COE). The focal point of this initiative lies in surmounting the challenges associated with establishing a streamlined infrastructure for integrating a hybrid renewable energy system (HRES). Our innovative system seamlessly combines solar photovoltaic technology with wind turbines and pumped hydroelectric technology for storage. The methodology used in this study involves a comprehensive assessment of the viability of a hybrid renewable energy system (HRES) designed to address the energy needs of a remote region using HOMER Pro. The operational plan for the HRES design is depicted in Fig. 1. This plan outlines the sequence of steps involved in the assessment, with each step contributing to a comprehensive evaluation of the hybrid system's viability.

The following steps outline the methodology employed:

2.1. Data collection and baseline evaluation

Data collection and a baseline evaluation were conducted to assess various aspects of the hamlet's energy requirements and existing energy infrastructure. This assessment also took into account ongoing and upcoming renewable energy initiatives in the region. The following key points were addressed:

- **Resources availability assessment:** evaluated the availability and sustainability of natural resources such as water, land, forests, and minerals to support various village activities, including agriculture, energy production, and livelihoods.
- **Energy Demand Evaluation:** assessed the existing energy infrastructure, including an evaluation of the reliability and capacity of the current power source from both financial and energy requirement perspectives.
- **Comprehensive Financial Assessment:** conducted a comprehensive examination of the village's financial considerations, including factors such as the cost of energy, cost projections, and overall economic viability.

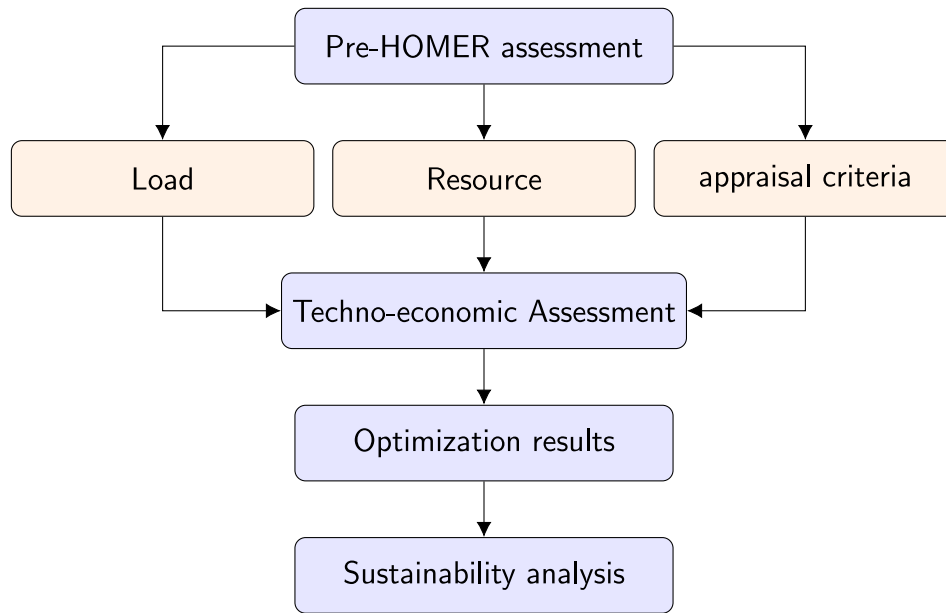


Fig. 1. HOMER design operational plan of the case study.

- Analysis of renewable energy initiatives: explored ongoing and planned renewable energy projects in the area, considering potential financial synergies, benefits related to the cost of energy, and alignment with energy demand.

Based on the collected data, a comprehensive understanding of the local financial and energy landscape, including implications related to the cost of energy and alignment with energy demand, was established. Compared to other software options like RETScreen and PVSyst, HOMER Pro is unique due to its intuitive interface, wide component selection, scenario analysis tools, geographic information systems (GIS) integration, and strong customer support. It strikes a balance between advanced features and user-friendliness, making it a favored solution for microgrid design and optimization.

2.2. Comprehensive viability analysis

To comprehensively assess the viability of the proposed hybrid renewable energy system from both technological and financial perspectives, we employed a multi-step approach:

- **Software Utilization:** We utilized HOMER Pro, a renowned microgrid design optimization tool developed at the National Renewable Energy Laboratory (NREL) and further enhanced by HOMER Energy. This software is widely recognized for its capabilities in optimizing microgrid designs (HOMER® Pro).
- **Economic Analysis:** Our study aimed to provide a comprehensive view of economic parameters, including cost-effectiveness, financial viability, and return on investment.
- **Environmental Impact Assessment:** An analysis of the system's environmental impact, with a focus on environmental sustainability, was conducted.

This analysis was conducted in consideration of Morocco's energy deficit and economic context, ensuring that the suggested hybrid system aligns with the country's energy goals.

2.3. Hybrid system configuration

Considering the local economy, energy requirements, and available renewable energy resources, the setup of the hybrid renewable energy

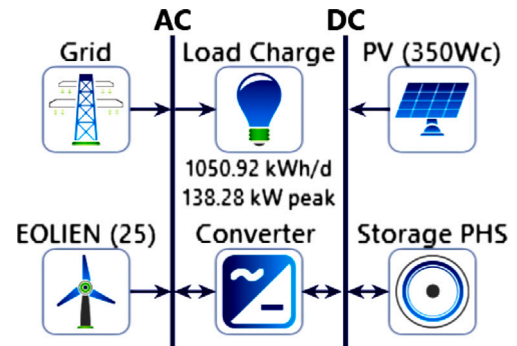


Fig. 2. Schematic layout of study.

system was determined. This process involved selecting suitable components, including pumped hydro energy storage (PHES or PHS), solar PV systems, wind turbines, and grid power to supply the system's energy demands or utilize surplus production, as shown in Fig. 2. Additionally, the 'optimal scale' for each power production technology was defined to ensure efficient energy generation.

2.4. Studied village

Tazarine is a rural community in Zagora Province, in the Draa-Tafilalet region of Morocco. Geographically, approximately located at 30°46'20" N; 5°32'45" W, with an altitude of 1058 meters above sea level. In terms of climate, Tazarine experiences a dry and hot desert climate when compared to the surrounding regions. As per the population estimate from 2014, the town's population is approximately 15,168 individuals, as reported on the data portal (Haut Commissariat au Plan, 2016). The village already has access to the grid, but the current power source lacks reliability in ensuring continuous electricity supply. Enhancing reliability is achievable through harnessing the advantages of readily available renewable energy sources. Our research concentrated on specific tribes in Tazarine (namely: Ait-Rebaa, Ait Imnasef, and Ait Baha), addressing the recurring power outages caused by the inadequate capacity of the electrical grid to supply energy to all town residents. The Fig. 3 shows the location of Tazarine.



Fig. 3. Location of Tazarine.

3. Hybrid energy system modeling

In this section, we start by explaining the mathematical and economic frameworks for each subsystem. Then, we provide a detailed overview of the solution methodology, which includes a comprehensive formulation of the proposed method, and optimization objectives.

3.1. PV system

The power generated (output) of the photovoltaic (PV) system is determined by Eq. (1) given by HOMER® Pro:

$$P_{PV} = Y_{PV} \times f_{PV} \times \left(\frac{G_{PV}}{G_{T,STC}} \right) \times \left[1 + \beta_p \times (T_c - T_{c,STC}) \right] \quad (1)$$

where:

- P_{PV} : PV Array Power Output [kW]
- Y_{PV} : Rated capacity of the PV array, power output under STC, [kW]
- f_{PV} : PV derating factor [%]
- G_{PV} : Incident radiation on the PV array in time step, [kW/m²]
- G_{STC} : Incident radiation at standard test conditions [1 kW/m²]
- β_p : Temperature coefficient of power [%/°C]
- T_c : PV cell temperature in the current time step [°C]
- $T_{c,STC}$: PV cell temperature under standard test conditions [25 °C]

The energy produced by the PV system over a certain period is given by the equation:

$$E_{PV} = P_{PV} \times t \quad (2)$$

where:

- E_{PV} : Energy produced by the PV system (in watt-hours or kilowatt-hours)
- P_{PV} : Power generated by the PV system (calculated using Eq. (1))
- t : Time period over which the energy is generated (in hours)

3.2. Wind turbine system

Wind turbines are crucial for clean energy generation, operating efficiently based on fluctuating wind speeds. Safety measures are implemented to align with wind speed variations, ensuring optimal performance. Therefore, wind velocity at the specified hub height (U_{hub}) and the associated power output (P_{wout}) are computed using the Eqs. (3), El-Houari et al. (2020):

$$U_{hub} = U_{anem} \left(\frac{Z_{hub}}{Z_{anem}} \right)^\alpha \quad (3)$$

where:

- U_{hub} : the wind speed at the hub height of the wind turbine [m/s]
- U_{anem} : the wind speed at anemometer height [m/s]
- Z_{hub} : the hub height of the wind turbine [m].
- Z_{anem} : the anemometer height [m]
- (α) : the power law exponent

Wind power density, a key indicator of wind potential, is proportional to the average cube of wind speed and relies on the frequency distribution of wind speeds. Power curves detail wind turbine performance in standard conditions. HOMER adjusts predicted power using the air density ratio for real-world conditions as specified in the subsequent equation (4):

$$P_{wt} = \left(\frac{\rho_{ref}}{\rho_0} \right) \cdot P_{wt,STP} \quad (4)$$

where:

- P_{wt} : the adjusted power,
- ρ_{ref} : the reference air density,
- ρ_0 : the standard air density,
- $P_{wt,STP}$: the power at standard temperature and pressure (STP).

The power generated by the wind turbine is given by Eq. (5), Yang et al. (2023):

$$P_{wind} = \frac{1}{2} \times \rho \times A \times v^3 \times C_p \quad (5)$$

where:

- P_{wind} : Power generated by the wind turbine (in watts),
- ρ : Air density (in Kg/m³),
- A : Swept area of the wind turbine blades (in m²),
- v : Wind speed (in m/s),
- C_p : Power coefficient, representing the efficiency of the wind turbine.

The energy produced by the wind turbine over a certain period is given by the equation:

$$E_{wind} = P_{wind} \times t \quad (6)$$

where:

- E_{wind} : Energy produced by the wind turbine (in watt-hours or kWh)
- P_{wind} : Power generated by the wind turbine (calculated using Eq. (6))
- t : time over which the energy is generated (in hours).

3.3. Storage system

Pumped hydro storage entails storing energy by pumping water uphill during periods of surplus electricity, and subsequently releasing it downhill to generate power during times of high demand. Furthermore, the researchers utilized a specific equation (7), to assess the available energy potential (Fan et al., 2020).

$$E_{(energy)} = \eta \times \rho_{(w)} \times g \times h_{(head)} \times V_{(reserv)} \quad (7)$$

where:

- $E_{(energy)}$: The energy stored in joules can be converted to kilowatt-hours by dividing it by 3.6×10^6 .
- η : the efficiency of energy conversion and must take into account factors such as turbine efficiency, generator efficiency, and hydrodynamic losses.
- $\rho_{(water)}$: the typical density of water, usually about 1000 kg/m³.
- g : the value of the acceleration due to gravity = 9.81 m/s²
- $h_{(head)}$: the head height in meters.
- $V_{(resrv)}$: the volume of the reservoir in m³.

3.4. Inverter system

The power output in AC form and energy delivered in AC form are given by HOMER in the following Eqs. (8), (9):

$$P_{AC} = P_{PV} \times \eta_{\text{converter}} \quad (8)$$

and

$$E_{AC} = P_{AC} \times t \quad (9)$$

where:

P_{AC} : Power output in AC form (in watts),

P_{PV} : Power generated by the PV system (calculated using Eq. (2)),

$\eta_{\text{converter}}$: Efficiency of the DC/AC converter,

E_{AC} : Energy delivered in AC form (in watt-hours or kilowatt-hours),

t : Period over which the energy is delivered (in hours).

3.5. Economic model for analysis

The results of the cost analysis, derived from HOMER Pro, provide valuable insights into the financial feasibility of the proposed energy system. Through a meticulous examination of diverse economic parameters, HOMER Pro offers a comprehensive overview of the project's financial landscape. This encompasses the computation of capital costs, net present cost (NPC), operational and maintenance (O&M) expenses, and the levelized cost of energy (LCOE). These findings are crucial in the decision-making process, allowing stakeholders to thoroughly evaluate the economic viability of the energy project. The following subsection displays the cost analysis in detail.

3.5.1. Levelized net present cost

Net Present Value: The total net present cost (NPC) of a system, calculated by HOMER, represents the present value of lifetime costs minus the present value of revenue. It includes capital, replacement, O&M, fuel, and grid-related expenses. The NPC is determined by summing discounted cash flows across each project year, providing a concise assessment of the system's economic viability (Elkadeem et al., 2019). The NPC value is calculated by HOMER using the following equation (10):

$$NPC_{(\$)} = \frac{TAC_{(\$/\text{yr})}}{CRF} \quad (10)$$

where:

- TAC is the Total Annualized Cost in (\$/year)
- CRF is the capital recovery factor, determined by (11) as follows:

$$CRF_{(i,n,N)} = \frac{i(1+i_n)^N}{(1+i_n)^N - 1} \quad (11)$$

- N is the lifetime of the project,
- f is the rate of nominal expected inflation,
- i' is the nominal discount rate,
- i is the rate of real discount, determined by (12) as follows:

$$i = \frac{i' - f}{1 + f} \quad (12)$$

3.5.2. Levelized cost of energy

The Levelized Cost of Energy/Cost of Energy (LCOE/COE) is a crucial economic indicator for hybrid system analysis; it represents the average cost per kilowatt-hour (kWh) of usable electric power derived from the system.

The COE is calculated by HOMER using the following (13):

$$COE_{(\$/\text{kWh})} = \frac{C_{\text{ann} \cdot \text{total}}}{E_{\text{ser}}} \quad (13)$$

where:

Table 1

Solar radiation data.

Month	GHI average in kWh/m ² /Month	GHI average in kWh/m ² /Day	Cleanness index
Jan	199.45	6.43	1.11
Feb	197.40	7.05	1.00
Mar	228.37	7.37	0.85
Apr	227.43	7.58	0.74
May	224.47	7.24	0.65
Jun	217.33	7.24	0.63
Jul	219.09	7.07	0.63
Aug	217.70	7.02	0.67
Sep	204.86	6.83	0.75
Oct	204.61	6.60	0.88
Nov	188.67	6.29	1.04
Dec	192.15	6.20	1.15

- $C_{\text{ann} \cdot \text{total}}$ is total annualized cost for the system [\$/yr],
- E_{serv} is the total energy load served [kWh/yr].

4. Assessment of village resources and energy demand

4.1. Resources assessment for the region of Tazarine

Tazarine town experiences a significant potential for renewable energy sources, particularly solar and high wind speeds, attributed to its geographical features, including mountainous terrain, desert surroundings, and the presence of an annual valley passage. Additionally, a dam is located in the vicinity. In response to these conditions, we have put forward a proposal for a pumping system, and we can improve this system with Grid, as we are constrained to a 25-kW supply from it. If there is excess production, we plan to feed it back to the network or supply the village, potentially generating revenue through the sale of surplus energy. Currently, there are restrictions on selling energy in Morocco at low voltage (LesEchos, 2021). In the event of a surplus, we aim to distribute it among other villages to sell it to facilities nearby or store it in the storage of PHES. The suggested HRES in the current project consists of a photovoltaic (PV), wind turbine, and pumped hydro energy storage (PHES or PHS). As a result, the assessment of the input storage basin and weather data is provided in the following subsections:

4.1.1. Solar irradiation

The solar radiation data for the Tazarine village location is retrieved from the "Photovoltaic Geographical Information System Database" (PVGIS, 2022). We insert the data given by PVGIS and use it in the program Homer Pro. We utilize solar Global Horizontal Irradiation (GHI) resources to estimate the electricity production from flat photovoltaic panels. The data demonstrates that the region possesses substantial solar energy potential and is capable of generating a significant amount of electricity through photovoltaic (PV) panels.

Table 1 summarizes the database of the average GHI and Cleanness Index. Figs. 4 and 5 showcase the average solar irradiation profile by day and month in kilowatt-hours per square meter per month (kWh/m²/month) and per day (kWh/m²/day). Additionally, they display the cleanness index and daily radiation generated by HOMER for the examined site. The annual average solar Global Horizontal Irradiance (GHI) is 6.91 kWh/m²/day.

4.1.2. Wind speed resources

Wind energy harnesses the motion of air as it passes through wind turbines to generate electricity. Nevertheless, the variability of wind as an energy source necessitates an evaluation of its potential at a specific location to optimize its efficiency (Sinha and Chandel, 2015). Numerous studies have thoroughly examined hybrid energy systems,

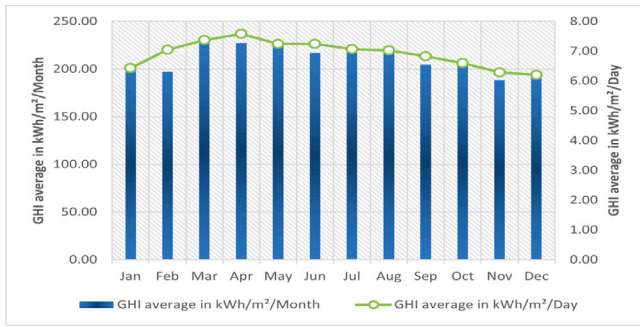


Fig. 4. Monthly average Solar GHI of the village.

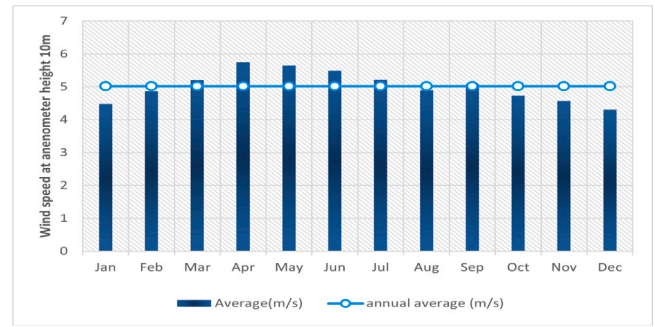


Fig. 6. Monthly average Wind Speed Data of the village.

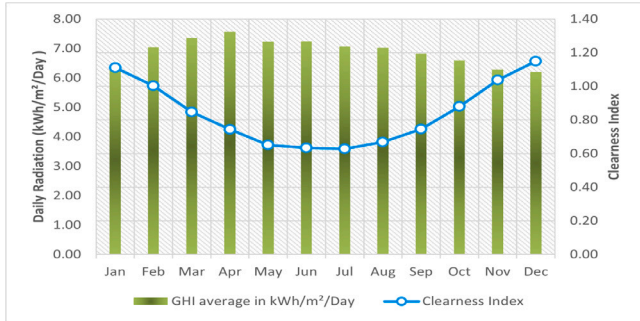


Fig. 5. GHI average in kWh/m²/Day with Clearness Index of the village.

Table 2

Wind speed data.

Month	Average wind speed (m/s)
Jan	4.48
Feb	4.87
Mar	5.21
Apr	5.75
May	5.65
Jun	5.5
Jul	5.22
Aug	4.9
Sep	5.09
Oct	4.73
Nov	4.58
Dec	4.31
Annual Average	5.02

Table 3

Temperature data.

Month	Temperature maximal (°C)	Temperature minimum(°C)	Monthly average temperature (°C)
Jan	17	3	10
Feb	20	5	12.5
Mar	24	8	16
Apr	28	11	19.5
May	32	15	23.5
Jun	37	20	28.5
Jul	41	23	32
Aug	40	23	31.5
Sep	35	20	27.5
Oct	29	15	22
Nov	22	9	15.5
Dec	18	5	11.5
Annual Average	28.58	13.08	20.83

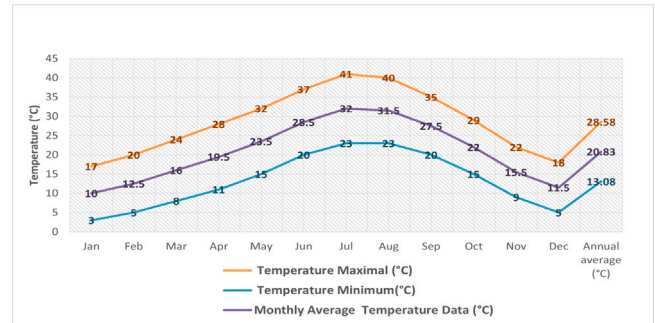


Fig. 7. Monthly temperature of the village.

highlighting the reliability and efficiency of wind power across varied global landscapes (Al-Ghussain et al., 2021; Bakić et al., 2012).

The prevailing wind patterns in Tazarine are typically characterized by the dominance of the dry and hot winds originating from the Sahara Desert, which arrived at an average of 6.03 m/s at an altitude of 100 m. Data for 10% windiest areas given by Global Wind Atlas (GWA) These winds, known as the “Sirocco” or “Chergui”, blow from the southeast and can bring elevated temperatures and low humidity levels to the region.

Our particular inquiry is assessing the wind speeds specific to the village of Tazarine, adding valuable insights to the broader understanding of renewable energy integration in distinct regions.

Table 2 illustrates the wind speed profile, obtained and compiled by the “NASA Prediction of Worldwide Energy Resource (Power) Database” using HOMER (HOMER® Pro) for the studied area. Fig. 6 depicts the annual average wind speed for this site, recorded at 5.02 m/s at an anemometer height (m) of about 10 m.

4.1.3. Climatic conditions of the village

The ambient temperature should also be taken into account when assessing the solar energy potential because it has an impact on the performance of PV modules. Table 3 summarizes the data provided by the “Meteoblue Climate Database” (Meteoblue, 2022) used to retrieve the average monthly temperatures for the selected village. Additionally, Fig. 7 displays the “Solar Energy Database” with an annual average temperature of 20 °C.

4.1.4. Pumped hydro energy resource

Pumped hydro-energy storage (PHES or PHS) is a proven technique for energy storage that harnesses the inherent potential energy of water (Ma et al., 2014). Typically employed in large-scale contexts, as detailed in previous sections, recent research endeavors are delving into its adaptability for smaller-scale applications. A multitude of case studies are actively pursuing the optimization of hybrid systems that seamlessly integrate renewable energy and PHES. These versatile applications encompass a wide spectrum of scenarios, ranging from urban settings and small communities with grid connectivity to remote

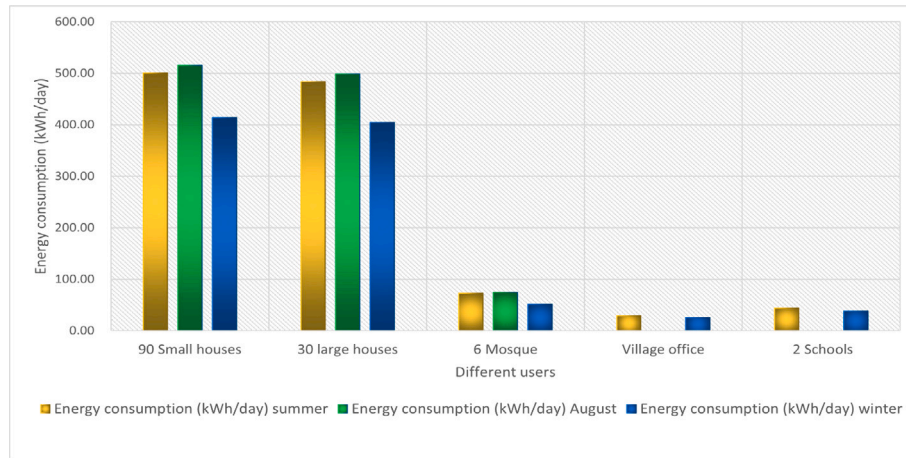


Fig. 8. Daily average load demand profile of summer for the end users.

Table 4

Energy demand for different users in the Village.

Different end users covered by the case study	Energy consumption (kWh/day)		
	summer	August	winter
90 Small houses	500.40	515.41	414.00
30 large houses	484.20	498.73	404.70
6 Mosque	72.60	74.78	51.00
Village office	29.44	0.00	25.24
2 Schools	43.68	0.00	38.58
Total demand of Energy	1130.32	1088.92	933.52
Annual Average of Energy (kWh/d)	1050.92		

rural areas and the inventive repurposing of abandoned surface mines. In essence, PHES stands at the forefront of energy storage solutions, offering a dynamic and promising approach for both established and emerging energy needs.

4.2. Load assessment

In contrast to urban residents, those living in rural areas generally have considerably reduced energy demands. Tazarine, situated in Morocco, will be the focus of our energy estimation. Our article will specifically concentrate on certain tribes in Tazarine, addressing their challenges with electricity scarcity and occasional power interruptions, as we have cited before. Therefore, we will focus on these tribes, which comprise ninety smaller houses, thirty large residences, six mosques, a village office, and two schools, as shown in Table 4. This estimation will encompass household loads (such as lighting, ovens, air conditioners, etc.) as well as community loads. Energy requirements in the area are categorized into two seasons, specifically during the summer period. (from April to September) into winter (from October to March) and We incorporated the data from August into our analysis, recognizing it as a significant holiday period for school, as the consumed energy ranges from 515.41 kWh/day as a peak to 25.24 kWh/day as a minimum load, with an annual average load of 1050.92 kWh/day, respectively, outlined in Fig. 8, and the peak of load charge is about 138.28 kW.

4.3. Detailed description of all components

This hybrid system's software optimized each scenario generated through the various micro-grid configurations to satisfy the village's demand load at the lowest possible cost of energy. Components of the study system are shown in Fig. 9.

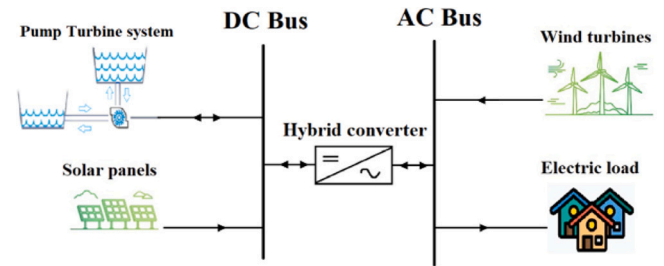


Fig. 9. Configuration of the studied system.

Table 5

Mechanical characteristics & specifications of photovoltaic.

Item	Unit	Specification
Cell Type	–	Mono-crystalline
Product Dimensions	(mm)	1956 × 992 × 40
Maximum Power Pmax	(Watts)	350
Maximum Amperage Imp	(A)	8.94
Maximum Voltage Vmp	(Volts)	39.10
Open-circuit Voltage	(Voc)	47.50
Short-circuit Current	(Isc)	9.88
Module Efficiency STC	(%)	18.04%
Operating Temperature	(°C)	–40 °C + 85 °C
Power tolerance	(%)	0 +3%
Nominal operating cell temperature (NOCT)	(°C)	45 ± 2 °C
Panel slope	(degrees)	31°

4.3.1. Solar PV system

The photovoltaic (PV) system harnessed in this project yields electricity in the form of direct current (DC). This versatile energy can be efficiently directed toward storage within the system or seamlessly transformed into alternating current (AC), catering to the dynamic power requirements of the load. This dual functionality not only ensures optimal energy utilization but also underscores the adaptability of our renewable energy infrastructure to diverse applications and electricity consumption needs. The type of panels used in the simulation is that of Jinko Solar Eagle JKM350M-72-V of 350 Wc, connected and oriented to the south at 31°. The characteristics and specifications of the PV array system carried on the DC bus are displayed in Table 5.

4.3.2. Wind turbine

The wind turbine investigated in this study is of the horizontal axis type, generating a rated 25 kW of AC. Technical details are outlined in Table 6, and its operational longevity spans 25 years.



Fig. 10. Proposed installation of the station of PHS for Upper and Lower reservoir.

Table 6

Mechanical characteristics & specifications of wind turbine.

Characteristic	Specification
Model	Eocycle EOX S-16
Design life	30 years
Rated power	25.0 kW
Rated wind speed	7.5 m/s
Cut-in wind speed	2.75 m/s (9.9 km/h)
Cut-out wind speed	20 m/s (72 km/h)
Tipspeed	59 m/s
Power density 1	199.8 W/m ²
Wind class	IEC Class IIIA
Power density 2	5.0 m ² /kW
Hub height	18/ 24/ m

Table 7

Main characteristics of the pumping stations.

Parameter	Value
Generic Hydro turbine	pumped-hydro 245 kWh
Nominal head	120 m
Nominal Flow rate	0.5 m ³ /s
Volume of our reservoir	4500 (m ³)(100 × 20 × 2.25)
Hydraulic pump efficiency	90%
Number of Discharge Hours	12 h
Flow rate of discharge (m ³ /s)	0.00104
Flow rate of charge (m ³ /s) at 90%	0.0759
The round trip efficiency of PHS	0.81 (81%)
The max Capacity (Ah)	5518.125
Lifetime	30 Years

4.3.3. Pumped hydro storage

A system for storing energy, known as Pumped Hydro, has been developed, utilizing water reservoirs positioned at different altitudes. During periods of surplus energy, water is transferred from a lower reservoir to a higher one, thereby creating potential energy (Petrolese et al., 2019). The study modeled the PHES station using the newly introduced 245 kWh pumped-hydro component from the HOMER library. This system can store and discharge 1000 m³ of water over 12 h (HOMER[®] Pro). Table 7 shows the characteristics of the pumping station. Fig. 10 presents the proposed position of installation of the station this investigation specifically focused on two vital components: turbines positioned before the treatment phase and turbines positioned after the treatment process (Michel Dubas, 2009).

4.3.4. Power converter

The inverter (converter) used is the ABB string inverter TRIO-20.0-TL-OUTD type of 20 kW, which costs 5454\$/kW. The choice of the

Table 8

Mechanical characteristics of converter.

Type code	Specification
Nominal output power	20000 W (20 kW)
Rated grid AC voltage	480 V
Absolute maximum voltage (Vmax)	1000 V
Grid connection type	3Ø/4 W + Ground
Total harmonic distortion at rated power	< 3%
Efficiency (Max/CEC)	98.2%/97.5%
Extended warranty	15–20 years
Ambient operating temperature range	(−30 °C to +60 °C)

inverter must be compatible with the other source of energy that we want to ensure. The other properties are presented in Table 8:

5. Results and discussion

This study designs and thoroughly examines a hybrid system to meet the energy demand of the site. Using HOMER, the optimization provides insights into system performance and sensitivity, identifying the optimal configuration of our proposed system used in HOMER Pro depicted in Fig. 2 which combines PV–wind–pumped hydro storage. The comparative analysis involves examining variations in economic and technical factors for each system being investigated. The details are considered in the following subsection:

5.1. Optimization results

The optimization scenario highlights a unique synthesis of outcomes for each component of the study. The ideal system, which encompasses all components such as PV–Wind turbine–PHS and converter, provides detailed specifications for the configuration of each element. The various options explored in the simulation's search space are presented in Table 9.

The outcomes of the HOMER simulation and optimization procedures indicate that among the four scenarios, the system configurations were ranked from the most to the least economically profitable. The hybrid energy systems are classified as 1-PV/Wind/PHS, 2-PV/Wind, 3-PV/PHS, and 4-PV, respectively, as shown in Table 9.

As discerned from the analyses conducted, the initial proposition, comprising PV/Wind/PHS, emerges as the optimal recommendation based on the results obtained from Homer's program. The detailed examination of various energy scenarios uncovers unique energy requirements for each proposed configuration, as depicted in Table 9.

- In the first scenario, marked by the integration of PV/Wind/PHS, the cumulative solar electrical energy output reaches 101 kW.

Table 9

Overview of optimization outcomes for various configurations of our HES.

Architecture	1- PV/Wind/PHS	2- PV/Wind	3-PV/PHS	4- PV
PV (kW)	101	244	207	1478
Wind (Unit)	4	4	0	0
PHS (Unit)	1	0	3	0
Converter (kW)	13.9	30.8	32.5	72.7
NPC (\$)	262,596	408,232	648,911	1.47M
COE (\$/kWh)	0.038	0.058	0.130	0.289
O&M (\$/yr)	3,064	4,448	18,391	26,817
Capital Cost (\$)	212,393	329,647	398,361	1.08M
Electric production (kWh/yr)	613,145	883,247	503,794	2,859,210
Excess production (kWh/yr)	67,781	323,454	77,549	2,460,626

To support this system, four wind turbine units are considered essential, complemented by one unit of pumped hydro storage. Moreover, the converter component in this scenario operates with a capacity of 13.9 kW.

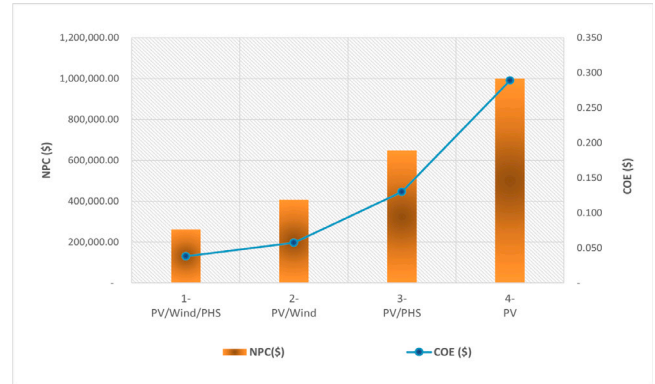
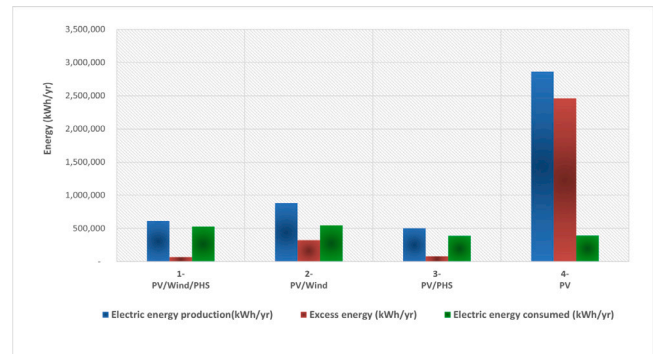
- Scenario 2, exclusively focused on PV/Wind, necessitates a more substantial energy infrastructure. A total of 244 kW of solar electrical energy is vital, along with four wind turbine units. The converter, pivotal for the energy conversion process, maintains a capacity of 30.8 kW to meet the demands of this configuration.
- In Scenario 3 (PV/PHS), which combines PV solar with pumped hydro storage, the analysis reveals a distinct energy profile. Approximately 207 kW of solar electrical energy is considered necessary, accompanied by three units of pumped hydro storage. The converter, integral to the energy conversion process, is suggested to have a rated capacity of 32.5 kW to optimize performance.
- Scenario 4 (PV), the final proposal, relies exclusively on solar photovoltaic technology. This configuration yields a substantial total energy output of 1478 kW. The nuanced characteristics of each scenario underscore the intricate interplay of diverse energy sources and the crucial role of converters in ensuring optimal performance across these distinct configurations.

Regarding the Net Present Cost (NPC), the first scenario stands out as the optimal choice, with an NPC reaching approximately \$262,596.2. Following closely is the second scenario, with an average NPC of approximately \$408,232.00. While the third scenario trails behind with an average NPC of approximately \$648,911.00. The fourth scenario stands out with a significant NPC, reaching approximately 1.47 million dollars as shown in Fig. 11.

Transitioning from NPC to the Cost of Energy (COE) or Levelized Cost of Energy (LCOE), the first scenario takes the lead with a rate of \$0.038. Following is the second scenario, with a rate of approximately \$0.058, while the third scenario sees an increase to \$0.130. Bringing up the rear is the fourth scenario, with the highest rate on the list at \$0.289. This hierarchy of rates sheds light on the cost-effectiveness of each scenario in terms of COE/LCOE, as shown in Fig. 11.

Concerning the Operational & Maintenance and capital costs, each scenario is outlined, respectively. Scenario 1 takes the lead with the lowest costs, amounting to \$3,064 and \$212,393. Following closely is scenario 2, with expenses totaling \$4,448 and \$329,647. Scenario 3 falls behind, with costs reaching \$18,391 and \$398,361. Lastly, scenario 4, classified as the larger type, incurs expenses of \$26,817 and \$1.08M, as illustrated in Table 9.

Regarding the total electrical energy produced, surplus electrical energy, and electrical consumed energy measured in kWh/year as shown in Fig. 12, distributed respectively as follows: In scenario 1, the combined energy generation, excess production, and electrical consumption amount to 613,145 kWh/year, 67,781 kWh/year, and 530,193 kWh/year, respectively. Scenario 2 follows with about 883,247 kWh/year, 323,454 kWh/year, and 548,557 kWh/year. Progressing to scenario 3, the values are 503,794 kWh/year, 77,549 kWh/year, and 3,387,112 kWh/year. Lastly, scenario 4 registers a total

**Fig. 11.** NPC and COE for all configuration.**Fig. 12.** Electric energy produced/consumed & Excess energy (kWh/yr).

of 2,859,21 kWh/year, 2,460,626 kWh/year, and 393,016 kWh/year, contingent on the pricing model.

5.2. Economical analysis

In this subsection, we will encapsulate the optimal scenario recommended by the program, which incorporates PV/wind/PHS. This choice proves to be the most efficient and fitting for supplying electricity to the village, enabling it to achieve energy self-sufficiency. Further details will be explored, specifically the costs associated with each component, derived from the obtained results.

The Cost Of Energy (COE) for our optimal system is approximately \$0.03831, making it the most cost-effective option compared to others. This optimization was achieved through the implementation of the best architecture proposed or fine-tuned by Homer Pro.

The summary NPC and annualized financial overview of the diverse elements in the hybrid PV/Wind/PHS encompassing the wind turbine,

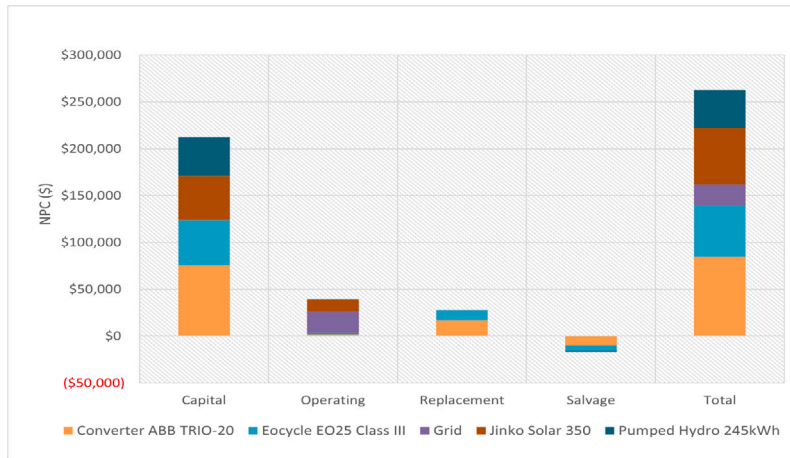


Fig. 13. Summary Net Present Cost for all components.

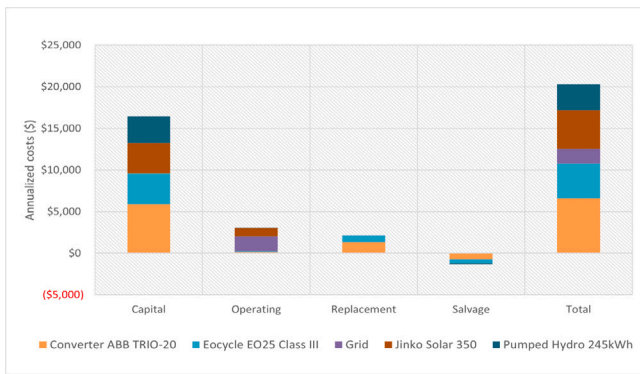


Fig. 14. Annualized costs Summary for all components.

PV array, power hydro storage, power converters, and grid, is depicted in Figs. 13 and 14.

The distribution of capital, replacement, O&M, and salvage costs for components by cost type, Tables 10 and 11 provide a detailed overview of the technical and financial characteristics of every part of our system under study.

Table 10, displaying the Summary Net Present Cost, illuminates the financial landscape. Of particular note is that the converter component incurs the highest cost, peaking at approximately \$84,825 due to its expensive technology, while the PHS component appears to have the lowest cost at \$40,674, providing a favorable comparison to other system elements.

Turning to Table 11, the Summary Annualized Cost table offers a detailed perspective on recurring financial considerations. Here, the PHS component stands out for its fiscal prudence, showcasing a minimal total cost of \$3,146. In contrast, the converter component records the highest annualized cost at \$6,562. For a comprehensive overview, Table 12 consolidates the total costs for all components, expressing them in terms of Net Present Cost (NPC) and annualized values. The total summary NPC of the system and the total annualized summary cost are \$262,596 and \$20,313, respectively.

6. Conclusion

This study highlights the critical role of hybrid renewable energy systems, particularly in desert regions and isolated villages, in offering vital solutions for addressing power outages and delivering clean energy. By integrating PV solar, wind turbines, and Pumped Hydro Storage (PHS), the research demonstrates the effectiveness of such

Table 10

Summary Net Present Cost (NPC) for the studied system.

Name	Capital(\$)	Operating (\$)	Replacement(\$)	Salvage(\$)
Converter	\$75,663	\$1,793	\$16,885	(\$9,516)
Wind	\$48,327	\$1,551	\$10,808	(\$6,091)
Grid	\$0.00	\$22,575	\$0.00	\$0.00
PV solar	\$46,890	\$13,036	\$0.00	\$0.00
PHS	\$41,513	\$652.84	\$0.00	(\$1,492)
System	\$212,393	\$39,609	\$27,693	(\$17,099)

Table 11

Summary annualized cost for the studied system.

Name	Capital(\$)	Operating (\$)	Replacement(\$)	Salvage(\$)
Converter	\$5,853	\$138.73	\$1,306	(\$736.10)
Wind	\$3,738	\$120.00	\$836.04	(\$471.16)
Grid	\$0.00	\$1,746	\$0.00	\$0.00
PV solar	\$3,627	\$1,008	\$0.00	\$0.00
PHS	\$3,211	\$50.50	\$0.00	(\$115.39)
System	\$16,430	\$3,064	\$2,142	(\$1,323)

Table 12

Total summary NPC and annualized cost for all components.

Name	Total NPC (\$)	Total annualized (\$)
Converter	\$84,825	\$6,562
Wind	\$54,596	\$4,223
Grid	\$22,575	\$1,746
PV solar	\$59,926	\$4,636
PHS	\$40,674	\$3,146
System	\$262,596	\$20,313

systems in meeting the energy needs of Tazarine, a village in the southern Sahara region of Morocco. Despite recent droughts and global water scarcity, the study finds that Tazarine benefits from favorable climatic conditions and geographical features, making it well-suited for PV array installation and wind energy generation. HOMER software analysis identifies the PV/Wind/PHS hybrid energy system as the optimal and cost-effective solution, with significantly lower Levelized Cost of Energy (LCOE) and Net Present Cost (NPC), presented respectively as \$0.03831, and \$262,596 compared to other configurations. Operating and maintenance costs are also minimized, while the system achieves an impressive 86% efficiency in energy consumption versus production. The study's focus on technical, optimization, and economic aspects underscores the potential of hybrid systems to rival grid connections, offering practical remedies for addressing energy challenges in rural African areas rich in renewable resources.

- As a future direction, we plan to explore evaluating hybrid renewable energy systems by entrusting control systems with their management. With this approach, we aim to optimize how these systems operate, ensuring they generate, store, and distribute energy intelligently, especially in fluctuating conditions. Our goal is to enhance their efficiency and reliability, making them more dependable in practical scenarios. We believe this research can propel us toward smarter and greener energy solutions, setting us on a path toward a more sustainable future.

CRedit authorship contribution statement

Norddine Oubouch: Writing – original draft, Methodology, Investigation, Conceptualization. **Abdelbari Redouane:** Writing – review & editing, Validation, Supervision, Conceptualization. **Anouar Makhoukh:** Conceptualization. **Abdennebi El Hasnaoui:** Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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